LEARN WITH SACHI / EXAM KUPPI

BMIT Calculation Workbook

IT3090 / IE3060 · සියලු calculations එක තැනක
Formula sheet · Worked examples · Practice questions · Answer key

A–H	14	Full	Nov 3
WORKED SECTIONS	PRACTICE QUESTIONS	ANSWER KEY	EXAM

Calculators are allowed in the exam. Bring yours.

How to use this workbook

Part	කරන්නේ මොකක්ද
1. Formula sheet	හැම formula එකම එක පිටුවක. exam එකට කලින් මේක ආපහු බලන්න.
2. Worked examples (A to H)	Past paper ප්‍රශ්න වල සම්පූර්ණ working. මුලින් ඔයාම කරලා බලලා පස්සේ compare කරන්න.
3. Practice questions	Answers නැති ප්‍රශ්න 14ක්. Kuppi එකේදී හෝ තනියම කරන්න.
4. Answer key	Practice ප්‍රශ්න වල උත්තර සහ working. පරීක්ෂා කරන්න විතරයි, මුලින් බලන්න එපා.

EXAM

හැම calculation එකකම **Formula → Substitution → Answer (units)** ලියන්න. Ratios වලට **interpretation** අනිවාර්යයි. Cost / profit නොදුන්නොත් **assumption** ලියන්න.

1. Formula sheet

Topic	Formula	Note
Accounting equation	Assets = Liabilities + Capital | Capital = Assets − Liabilities | Liabilities = Assets − Capital	හැමවිටම tally
Profit on a sale	Profit = Selling price − Cost	Capital increases by profit
Gross profit rate	(Gross profit ÷ Net revenue) × 100	Higher is better
Net profit rate	(Net profit ÷ Net revenue) × 100	Higher is better
Current ratio	Current assets ÷ Current liabilities	About 2.0
Quick ratio	(Current assets − Inventory) ÷ Current liabilities	1.0 or more
Receivable days	(Closing trade receivables × 365) ÷ Revenue	Lower is better
Gearing ratio	(Debt ÷ Equity) × 100	Higher means higher risk
Inventory turnover *	Cost of goods sold ÷ Inventory (times)	Higher is faster
Total assets turnover *	Revenue ÷ Total assets (times)	Higher is better
Debt-to-equity *	Total debt ÷ Total equity	State which debt you used
Mark-up price	Selling price = Cost × (1 + mark-up %) | Cost = Selling price ÷ (1 + mark-up %)	Mark-up is on cost
Target cost	Target cost = Selling price − Required profit (or SP ÷ (1 + mark-up %))	Compare with current cost
Equilibrium	Set Qd = Qs, solve for P, substitute to find Q, check with both equations	Qd = a − bP ; Qs = c + dP
Surplus / shortage	Surplus = QS − QD (price above equilibrium) | Shortage = QD − QS (price below)	
Shifts	D up: P up, Q up | D down: P down, Q down | S up: P down, Q up | S down: P up, Q down	D: same direction; S: opposite
PPC opportunity cost	Fall in one good ÷ rise in the other good between two points	Increasing for a concave PPC
EPF / ETF	Employee EPF 8% · Employer EPF 12% · Employer ETF 3% (of basic salary)	Employer cost = salary + 15%

* Not in the lecture slides but asked in past papers (2022, 2024).

2. Worked examples

A. Accounting equation

METHOD

1. හැම transaction එකකටම බලපාන items 2ක් හොයන්න. 2. + / − දාන්න. 3. අන්තිමට totals ගණන් කරලා **Assets = Liabilities + Capital** පරීක්ෂා කරන්න. Profit වලට: විකුණපු මිල ඇතුළු assets වැඩි, inventory (cost) අඩු, capital එකට profit එකතු.

A1. Missing values (Lecture activity)

	Assets	Liabilities	Capital	Working
a	26,500	3,670	**22,830**	26,500 − 3,670
b	14,333	**4,828**	9,505	14,333 − 9,505
c	**17,150**	4,490	12,660	4,490 + 12,660
d	54,337	15,347	**38,990**	54,337 − 15,347
e	29,001	**1,446**	27,555	29,001 − 27,555
f	**9,560**	58	9,502	58 + 9,502

2019 MCQ: assets 8,000 and capital 6,000 gives liabilities **2,000**. Liabilities 5,000 and capital 1,000 gives assets **6,000**. Assets 50,000 and liabilities 10,000 gives capital **40,000**. Assets up 20,000 and liabilities up 12,000 gives capital up **8,000**.

A2. Nimal's phone accessories shop (10 transactions, with profit and drawings)

#	Transaction	ASSETS				LIABILITIES		CAPITAL
		Cash	Debtors	Inventory	Equipment	Creditors	Bank loan	Capital
1	Nimal invests Rs. 500,000 cash	+500,000						+500,000
2	Buys shelves & computer – cash	−120,000			+120,000			
3	Buys phone cases on credit			+200,000		+200,000		
4	Pays shop rent in cash	−30,000						−30,000
5	Sells cases for 90,000 cash (cost 60,000)	+90,000		−60,000				+30,000
6	Pays supplier 100,000	−100,000				−100,000		
7	Gets bank loan	+300,000					+300,000	
8	Sells cases 50,000 on credit (cost 35,000)		+50,000	−35,000				+15,000
9	Friend pays 20,000	+20,000	−20,000					
10	Nimal takes 10,000 for personal use	−10,000						−10,000
	TOTAL	**650,000**	**30,000**	**105,000**	**120,000**	**100,000**	**300,000**	**505,000**

Tally check: Assets 905,000 = Liabilities 400,000 + Capital 505,000 = 905,000 Tally OK

Capital 505,000 = 500,000 invested + profits (30,000 + 15,000) − rent 30,000 − drawings 10,000.

A3. 2022 Q3 · Sugath (30 marks)

Assumptions: sales "worth" Rs. 50,000 on credit are treated as income (capital up) because the cost is not given; goods taken for personal use reduce inventory and capital; the machine is Rs. 75,000 cash plus Rs. 75,000 on credit; cheques and loans go to Cash/Bank.

#	Transaction	ASSETS				LIABILITIES		CAPITAL
		Cash/Bank	Debtors	Inventory	Fixed assets	Creditors	Bank loan	Capital
1	Invested cash in business	+200,000						+200,000
2	Goods bought on credit			+100,000		+100,000		
3	Goods sold on credit to Nirmal (sales value)		+50,000					+50,000

4	Electricity expense paid in cash	−10,000						−10,000
5	Drawings – goods taken for personal use			−25,000				−25,000
6	Generator bought on credit (Abans)				+250,000	+250,000		
7	Bank loan obtained	+500,000					+500,000	
8	Nirmal paid 50% (25,000) by cheque	+25,000	−25,000					
9	Rent paid in cash	−40,000						−40,000
10	Additional capital invested	+100,000						+100,000
11	Goods sold on credit to Rukshala		+50,000					+50,000
12	Machine 150,000 – 75,000 cash, 75,000 credit (Singer)	−75,000			+150,000	+75,000		
13	Drawings – cash 25,000	−25,000						−25,000
14	Nirmal paid 25,000 cash	+25,000	−25,000					
15	Vehicle bought on credit (Allison)				+4,500,000	+4,500,000		
	TOTAL	**700,000**	**50,000**	**75,000**	**4,900,000**	**4,925,000**	**500,000**	**300,000**

Tally check: Assets 5,725,000 = Liabilities 5,425,000 + Capital 300,000 = 5,725,000 Tally OK

A4. 2023 Q3(b) · Nirmal (10 marks)

#	Transaction	ASSETS			LIABILITIES	CAPITAL
		Cash	Inventory	Machinery	Creditor (Kamal)	Capital
1	Nimal invested 100,000	+100,000				+100,000
2	Goods bought on credit from Kamal		+50,000		+50,000	
3	Machinery bought for cash	−15,000		+15,000		
4	Goods returned to Kamal		−5,000		−5,000	
5	Paid Kamal 2,000	−2,000			−2,000	
	TOTAL	**83,000**	**45,000**	**15,000**	**43,000**	**100,000**

Tally check: Assets 143,000 = Liabilities 43,000 + Capital 100,000 = 143,000 Tally OK

A5. 2023 ISE Q3 · Chirantha (30 marks)

Assumption: cash sales of Rs. 40,000 are income (cost not given).

#	Transaction	ASSETS			LIABILITIES	CAPITAL
		Cash/Bank	Inventory	Furniture/Computer	Creditors	Capital
1	Invested cash 1,200,000	+1,200,000				+1,200,000
2	Start-up expenses paid in cash	−12,000				−12,000
3	Goods bought on credit		+350,000		+350,000	
4	Cash sales (income)	+40,000				+40,000
5	Staff salary paid in cash	−50,000				−50,000
6	Paid a creditor by cheque	−50,000			−50,000	
7	Furniture on credit (Aruna Furniture)			+30,000	+30,000	
8	Rent paid in cash	−15,000				−15,000
9	Settled Aruna Furniture in cash	−30,000			−30,000	
10	Computer bought by cheque	−70,000		+70,000		
	TOTAL	**1,013,000**	**350,000**	**100,000**	**300,000**	**1,163,000**

Tally check: Assets 1,463,000 = Liabilities 300,000 + Capital 1,163,000 = 1,463,000 Tally OK

A6. 2024 Q4(a) · Theeraka (20 marks)

Assumption: office supplies bought for cash are treated as an asset (a swap); if treated as an expense, capital falls by 500 instead.

#	Transaction	ASSETS				LIABILITIES		CAPITAL
		Cash	Inventory	Equipment	Supplies	Payables	Bank loan	Capital
1	Owner invests 50,000 cash	+50,000						+50,000
2	Equipment bought for cash	−10,000		+10,000				
3	Inventory bought on credit		+8,000			+8,000		
4	Rent paid in cash	−2,000						−2,000
5	Sold inventory (cost 6,000) for 9,000 cash	+9,000	−6,000					+3,000
6	Paid suppliers 5,000	−5,000				−5,000		
7	Owner withdrawal (drawings)	−3,000						−3,000
8	Bank loan received	+20,000					+20,000	
9	Salaries paid in cash	−4,000						−4,000
10	Office supplies bought for cash	−500			+500			
	TOTAL	**54,500**	**2,000**	**10,000**	**500**	**3,000**	**20,000**	**44,000**

Tally check: Assets 67,000 = Liabilities 23,000 + Capital 44,000 = 67,000 Tally OK

B. Financial ratios

METHOD

1. Statements වලින් අංක ගන්න (වර්ෂ දෙකටම). 2. **Formula, working, answer** table එකක ලියන්න. 3. හැම ratio එකකටම **interpretation** (increase / decrease, තේරුම, හොඳද නරකද, ඇයි, යෝජනාව).

B1. 2019 Q3(a) · Sithumina Enterprise

Ratio	Working	Answer
Net profit rate	30,000 ÷ 250,000 × 100	**12%**
Current ratio	CA = 40,000 + 25,000 + 5,000 = 70,000; 70,000 ÷ 35,000	**2.0**
Quick ratio	(70,000 − 40,000) ÷ 35,000	**0.86**
Gross profit rate	100,000 ÷ 250,000 × 100	**40%**

B2. 2022 Q4 · Soveen Ltd (30 marks)

The paper's years are labelled inconsistently (2020/2021 and 2021/2022) and total assets 455 is a typo (285 + 160 = 445). Left column = Year 1, right column = Year 2. State this in the exam.

Ratio	Working	Year 1	Year 2
Gross profit margin	250/500 ; 300/480	50.0%	62.5%
Net profit margin	118/500 ; 172/480	23.6%	35.8%
Current ratio	160/115 ; 165/90	1.39	1.83
Quick ratio	(160−100)/115 ; (165−80)/90	0.52	0.94
Receivable days	50×365/500 ; 80×365/480	36.5	60.8
Total assets turnover	500/445 ; 480/415	1.12	1.16

Interpretation: profitability and liquidity improved; the main concern is receivable days rising from 36.5 to 60.8 (slower collection).

B3. 2023 Q3(c) · Lydex Company (10 marks)

Left column (CA 7,520,000; CL 4,010,000; inventory 3,600,000) treated as **this year**; right column (5,500,000; 2,980,000; 2,000,000) as **last year**.

Ratio	Working	Last year	This year
Current ratio	5,500/2,980 ; 7,520/4,010	1.85	1.88
Quick ratio	(5,500−2,000)/2,980 ; (7,520−3,600)/4,010	1.17	0.98

Interpretation: quick ratio fell below 1.0 because inventory grew; a liquidity warning.

B4. 2023 ISE Q4(b) · Typical Business Inc.

Ratio	Working	2008	2009
Current ratio	7,675/3,830 ; 8,815/4,030	2.00	2.19
Quick ratio	(7,675−2,725)/3,830 ; (8,815−3,450)/4,030	1.29	1.33
Gearing (short + long term notes)	(2,150+3,850)/5,730 ; (2,250+4,000)/6,820 × 100	104.7%	91.6%
Gearing (long-term notes only)	3,850/5,730 ; 4,000/6,820 × 100	67.2%	58.7%

B5. 2024 Q3 · Kottu Factory Ltd (Rs. '000)

Ratio	Working	Answer
Gross profit margin	150,000 ÷ 500,000 × 100	**30%**
Net profit margin	75,000 ÷ 500,000 × 100	**15%**
Current ratio	225,000 ÷ 75,000	**3.0**
Inventory turnover	350,000 ÷ 75,000 (closing inventory)	**4.67 times** (about 78 days)
Debt-to-equity	(100,000 + 25,000) ÷ 350,000 [or total liabilities 175,000 ÷ 350,000]	**0.36** [0.50]

Assessment: strong position (good margins, high liquidity, low gearing). Concerns: current ratio 3.0 may mean idle assets; inventory turnover could improve; no comparison year given.

B6. Kandy Bakes Ltd (worked from the kuppi)

Ratio	Working	Year 1	Year 2
Gross profit rate	800/2,000 ; 1,080/2,400	40.0%	45.0%
Net profit rate	300/2,000 ; 400/2,400	15.0%	16.7%
Current ratio	600/300 ; 800/500	2.00	1.60
Quick ratio	350/300 ; 500/500	1.17	1.00
Receivable days	200×365/2,000 ; 400×365/2,400	36.5	60.8
Gearing	400/1,000 ; 600/1,100	40.0%	54.5%

INTERPRETATION TEMPLATE

"The **[ratio]** has increased/decreased from **[X]** to **[Y]**. This means [what it measures]. This is good/bad because [reason]. The company should [suggestion]."

C. Mark-up pricing and target costing

C1. 2022 Q2 · Rosmee perfume

Step	Working	Answer
Unit cost	165 + (1,500,000 ÷ 50,000 = 30)	**Rs. 195**
Mark-up price (25%)	195 × 1.25	**Rs. 243.75**

Pricing strategy: **skimming** (new, exclusive, brand-conscious niche; perceived value Rs. 200 above competitor Rs. 175). Caution: mark-up price (243.75) is above perceived value (200).

C2. 2019 Q3(b) · technical device

Step	Working	Answer
Target cost	210 ÷ 1.40 = 150; × 5,000	**Rs. 150 / unit; Rs. 750,000**
Current cost	35,000 + 140 × 5,000 + 30,000	**Rs. 765,000** (Rs. 153 / unit)
Gap	765,000 − 750,000	**Rs. 15,000 above target**: not worth it on the current basis
Extra design 15,000	750,000 − (35,000 + 15,000) − 30,000 = 670,000; ÷ 5,000	**Rs. 134 / unit** maximum

C3. Phone cover (kuppi example)

SP 1,400; 2,000 units; mark-up 40%: target = 1,400 ÷ 1.4 = **Rs. 1,000** (Rs. 2,000,000). Current = 150,000 + 900 × 2,000 + 80,000 = **Rs. 2,030,000** (gap Rs. 30,000). Extra design 40,000: maximum manufacturing = (2,000,000 − 190,000 − 80,000) ÷ 2,000 = **Rs. 865 / unit**.

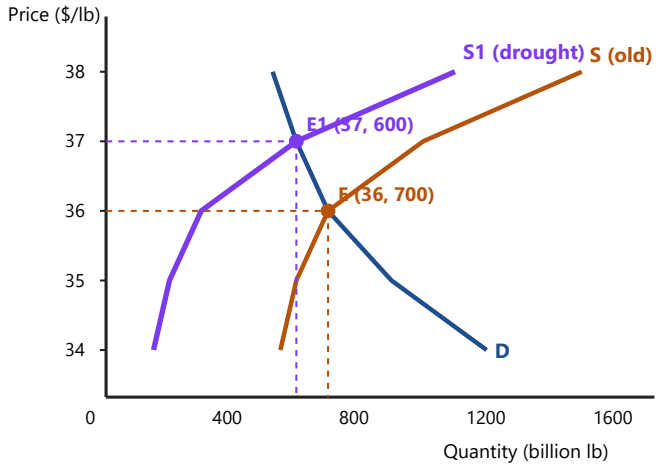
D. Equilibrium (demand and supply)

Paper	Equations	Working	Answer
2019 Q1(b)	Qd = 5000 − 100P ; Qs = 3000 + 300P	2000 = 400P	**P = 5, Q = 4,500**
2023 Q4(a)	D = 100 − 8P ; S = 28 + 5P	72 = 13P	**P = 5.54, Q = 55.7**
2023 ISE Q2(b)	D = 100 − 6P ; S = 28 + 3P	72 = 9P	**P = 8, Q = 52**
2023 Q4(b)	Qz = 150 − 5Pz + 2Y − 10Pw ; Y = 50, Pw = 6	150 − 5Pz + 100 − 60	**Qz = 190 − 5Pz**
Kuppi example	Qd = 120 − 4P ; Qs = 20 + 6P	100 = 10P	**P = 10, Q = 80**

D1. 2024 Q2(b) · world tea market (table)

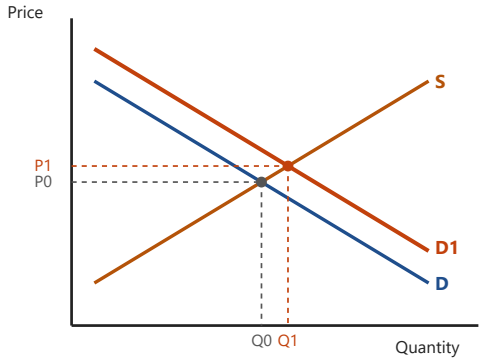
Price ($)	QS	QD	Condition	New QS (drought, −400)	New QS − QD
38	1,500	525	Surplus 975	1,100	+575
37	1,000	600	Surplus 400	600	**0 (new equilibrium)**
36	**700**	**700**	**Equilibrium**	**300**	**−400**
35	600	900	Shortage 300	200	−700
34	550	1,200	Shortage 650	150	−1,050

Original equilibrium **$36, 700**. After the drought: **$37, 600** (price up, quantity down).

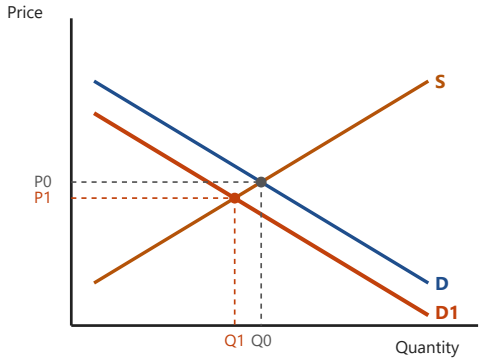


2024 Q2(b) – tea market: equilibrium E, then new equilibrium E1 after the drought

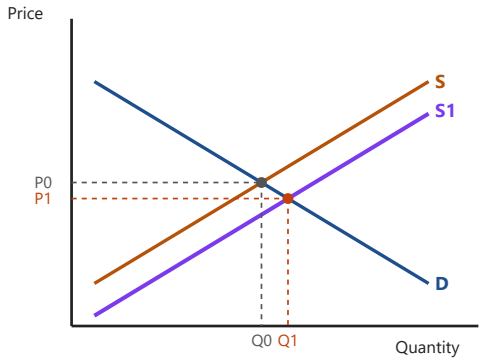
E. Shifts and diagrams



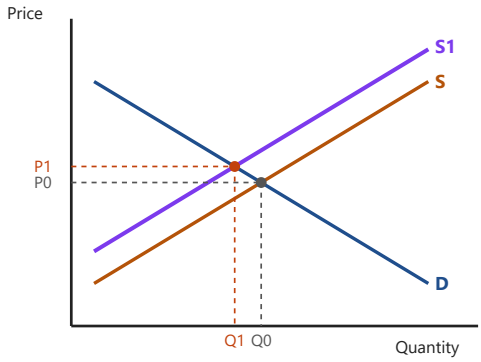
Demand ↑ (D → D1) → P ↑, Q ↑



Demand ↓ (D → D1) → P ↓, Q ↓



Supply ↑ (S → S1) → P ↓, Q ↑



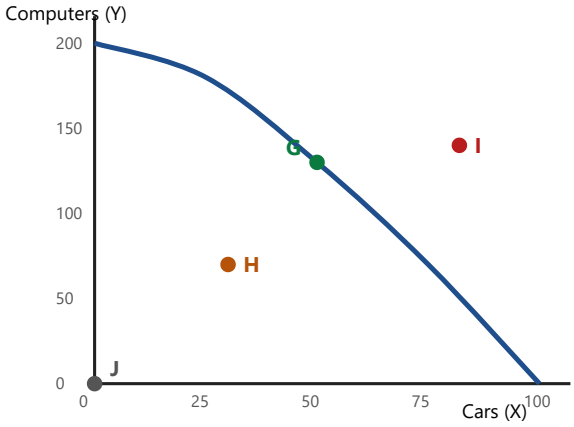
Supply ↓ (S → S1) → P ↑, Q ↓

Event	Curve	Shift	P	Q
"OL" T-shirt brand of the year	Demand	Right	Up	Up
Price of cotton rises	Supply	Left	Up	Down
People think rice makes them fat	Demand	Left	Down	Down
Less time to prepare meals (rice)	Demand	Left	Down	Down
Lecturer makes the IT textbook compulsory	Demand	Right	Up	Up
Printing costs fall	Supply	Right	Down	Up
Yoghurt: disease reduces milk / cheese price rises / expectation of higher price	Supply	Left	Up	Down
Yoghurt: new technology cuts costs	Supply	Right	Down	Up
Mirinda price rises Rs. 20 to 25 (35 to 30)	Movement along D: contraction (no shift)			

Fanta gets cheaper (for Mirinda)	Demand	Left	Down	Down

F. PPC and opportunity cost

Cars	0	25	50	75	100
Computers	200	180	130	70	0
Opportunity cost of the extra 25 cars		20	50	60	70



G = efficient (on the PPC) | **H** = inefficient (inside) | **I** = unattainable (outside) | **J** = origin (no production / no technology)

Increasing opportunity cost, so the PPC is concave. Points: on the curve = efficient; inside = inefficient; outside = unattainable; origin = no production.

G. EPF and ETF

Basic salary Rs. 100,000: employee EPF 8% = **8,000**; employer EPF 12% = **12,000**; employer ETF 3% = **3,000**. Cost to the employer = 100,000 + 12,000 + 3,000 = **115,000**.

H. Quick-fire calculation types (MCQ style)

#	Question	Working	Answer
1	Cost Rs. 400, mark-up 25%. Selling price?	400 × 1.25	**Rs. 500**
2	Selling price Rs. 630, mark-up 40%. Cost?	630 ÷ 1.40	**Rs. 450**
3	Qd = 300 − 10P, Qs = 60 + 20P. P and Q?	240 = 30P; Q = 300 − 80	**P = 8, Q = 220**
4	CA 90,000; inventory 30,000; CL 60,000. Current and quick ratio?	90/60 ; (90−30)/60	**1.5 ; 1.0**
5	Revenue 300,000; net profit 45,000. Net profit rate?	45/300 × 100	**15%**
6	Receivables 40,000; revenue 480,000. Receivable days?	40,000 × 365 ÷ 480,000	**30.4 days**
7	Debt 150,000; equity 300,000. Gearing?	150/300 × 100	**50%**
8	Assets up 50,000, liabilities down 10,000. Change in capital?	ΔA = ΔL + ΔC, so 50 = −10 + ΔC	**Up 60,000**
9	Own price of a good falls. Shift or movement?	Only price changed	**Movement along D (extension)**

3. Practice questions

RULES

කාලය: ප්‍රශ්නයකට විනාඩි 5 ත් 8 ත්. **Formula, working, answer** ලියන්න. Answer key එක බලන්නේ ඔයා ලියලා ඉවර වුණාම විතරයි. Total: 14 ප්‍රශ්න.

P1 · ACCOUNTING EQUATION — missing values

	Assets	Liabilities	Capital
a	48,000	12,500	?
b	?	7,200	21,800
c	65,000	?	40,000

P2 · ACCOUNTING EQUATION — Saman's repair shop

Record each transaction in a table (Cash, Equipment, Inventory | Payables, Loan | Capital) and show that the equation tallies at the end.

1. Saman started the business with cash Rs. 300,000.
2. Bought equipment for cash Rs. 80,000.
3. Bought inventory on credit Rs. 60,000.
4. Sold inventory costing Rs. 20,000 for cash Rs. 35,000.
5. Paid the supplier Rs. 30,000 in cash.
6. Took a bank loan of Rs. 100,000 (cash received).
7. Paid salaries Rs. 25,000 in cash.

State the totals of assets, liabilities and capital.

P3 · RATIOS — Ceylon Foods

Revenue Rs. 800,000; cost of goods sold Rs. 520,000; net profit Rs. 96,000; current assets Rs. 360,000 (including inventory Rs. 120,000 and trade receivables Rs. 100,000); current liabilities Rs. 180,000.
Calculate: gross profit rate, net profit rate, current ratio, quick ratio, receivable days and inventory turnover.

P4 · RATIOS AND INTERPRETATION — Two years

Rs.	Year 1	Year 2
Revenue	1,000,000	1,200,000
Gross profit	400,000	420,000
Net profit	150,000	120,000
Current assets	500,000	450,000
Inventory	200,000	250,000
Current liabilities	250,000	300,000
Debt	300,000	450,000
Equity	600,000	600,000

(a) Calculate gross profit rate, net profit rate, current ratio, quick ratio and gearing for both years.
(b) Interpret each change and give one recommendation to the management.

P5 · RECEIVABLE DAYS — using P4

Closing trade receivables were Rs. 80,000 (Year 1) and Rs. 180,000 (Year 2). Calculate the receivable days for both years and comment on the change.

P6 · MARK-UP PRICING — full costing

Direct materials Rs. 120 per unit; direct labour Rs. 40 per unit; fixed overheads Rs. 600,000 for 10,000 units. Mark-up 30%.
(a) Unit cost. (b) Mark-up selling price. (c) The customers' perceived value is Rs. 300 and a competitor sells at Rs. 260. Suggest whether the mark-up price is suitable.

P7 · TARGET COSTING — smart speaker

Selling price Rs. 1,800; required mark-up 50%; volume 4,000 units. Current costs: design Rs. 400,000; materials Rs. 1,000 per unit; other costs Rs. 600,000.
(a) Target cost per unit and in total. (b) Current total cost and the gap. (c) The maximum materials cost per unit that meets the target.

P8 · EQUILIBRIUM — equations

Qd = 200 − 5P and Qs = 40 + 3P.
(a) Equilibrium price and quantity. (b) What happens at P = 25? (c) What happens at P = 15? (d) Demand increases to Qd = 260 − 5P. Find the new equilibrium.

P9 · EQUILIBRIUM — market table

Price (Rs.)	50	60	70	80	90
Quantity demanded	900	700	500	300	100
Quantity supplied	100	300	500	700	900

(a) Equilibrium. (b) Surplus or shortage at Rs. 90 and at Rs. 50. (c) Supply increases by 400 at every price. Find the new equilibrium and say what happened to price and quantity.

P10 · SHIFTS — 5 events

For each event name the curve that shifts, the direction, and the effect on price and quantity.

a. The government gives a subsidy to rice farmers.
b. Incomes rise (laptops are a normal good).
c. The price of tea rises (effect on the market for coffee).
d. Floods destroy a large part of the vegetable crop.
e. A health report says sugary drinks are harmful.

P11 · PPC

rice and coconut

Rice (tonnes)	0	10	20	30	40
Coconut (thousand nuts)	100	90	70	40	0

(a) Opportunity cost of each extra 10 tonnes of rice. (b) Is the combination 20 tonnes and 80 thousand nuts attainable? (c) Is 20 tonnes and 50 thousand nuts efficient? Explain. (d) What does the shape of the PPC show?

P12 · EPF AND ETF

Basic monthly salary Rs. 120,000. Find the employee EPF, employer EPF, employer ETF, the employee's salary after EPF, and the total cost to the employer.

P13 · ACCOUNTING EQUATION

profit and drawings

Opening: cash Rs. 200,000, capital Rs. 200,000. Then: (1) bought inventory for Rs. 80,000 cash; (2) sold half of it for Rs. 70,000 cash; (3) paid rent Rs. 15,000 cash; (4) owner withdrew Rs. 10,000 cash for personal use. Find the closing assets, liabilities and capital.

P14 · MIXED QUICK-FIRE

7 parts

a. Cost Rs. 560, mark-up 25%. Selling price?
b. Selling price Rs. 975, mark-up 30%. Cost?
c. Qd = 400 − 8P and Qs = 40 + 10P. P and Q?
d. CA 120,000; inventory 45,000; CL 50,000. Current ratio and quick ratio?
e. Revenue 450,000; net profit 63,000. Net profit rate?
f. Receivables 54,000; revenue 600,000. Receivable days?
g. Debt 90,000; equity 360,000. Gearing?

4. Answer key

CAUTION **WAIT**

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P1

(a) Capital = 48,000 − 12,500 = **35,500** (b) Assets = 7,200 + 21,800 = **29,000** (c) Liabilities = 65,000 − 40,000 = **25,000**

P2 · Saman's repair shop

Item	Working	Closing
Cash	300,000 − 80,000 + 35,000 − 30,000 + 100,000 − 25,000	300,000
Equipment	80,000	80,000
Inventory	60,000 − 20,000	40,000
Total assets		**420,000**
Payables	60,000 − 30,000	30,000
Bank loan	100,000	100,000
Total liabilities		**130,000**
Capital	300,000 + profit 15,000 (35,000 − 20,000) − salaries 25,000	**290,000**

Tally: 130,000 + 290,000 = 420,000. OK.

P3 · Ceylon Foods

Ratio	Working	Answer
Gross profit rate	(800,000 − 520,000) ÷ 800,000	**35%**
Net profit rate	96,000 ÷ 800,000	**12%**
Current ratio	360,000 ÷ 180,000	**2.0**
Quick ratio	(360,000 − 120,000) ÷ 180,000	**1.33**
Receivable days	100,000 × 365 ÷ 800,000	**45.6 days**
Inventory turnover	520,000 ÷ 120,000	**4.33 times**

P4 · Two-year comparison

Ratio	Working	Year 1	Year 2
Gross profit rate	400/1,000 ; 420/1,200	40.0%	35.0%
Net profit rate	150/1,000 ; 120/1,200	15.0%	10.0%
Current ratio	500/250 ; 450/300	2.00	1.50
Quick ratio	(500−200)/250 ; (450−250)/300	1.20	0.67
Gearing	300/600 ; 450/600	50.0%	75.0%

Interpretation: revenue grew but both margins fell (higher costs or discounting), so profitability weakened. Liquidity worsened and the quick ratio dropped to 0.67, well below 1.0 (inventory rose while current assets fell). Gearing rose to 75%, so financial risk is higher. **Recommendation:** control costs and reduce inventory; avoid more borrowing until liquidity improves.

P5 · Receivable days

Year 1: 80,000 × 365 ÷ 1,000,000 = **29.2 days**. Year 2: 180,000 × 365 ÷ 1,200,000 = **54.75, about 54.8 days**. Customers now take nearly twice as long to pay (bad): cash is tied up, which also explains the weaker liquidity. Tighten credit terms, follow up debtors, offer early payment discounts.

P6 · Mark-up pricing

(a) Fixed overhead per unit = 600,000 ÷ 10,000 = 60; unit cost = 120 + 40 + 60 = **Rs. 220**. (b) 220 × 1.30 = **Rs. 286**. (c) 286 is below the perceived value (300) but above the competitor (260). It is acceptable if the product has clear extra value (quality, brand); otherwise

sales may be lost to the competitor, so consider a price nearer Rs. 260 to 280 or add value.

P7 · Target costing

(a) Target per unit = 1,800 ÷ 1.50 = **Rs. 1,200**; total = 1,200 × 4,000 = **Rs. 4,800,000**. (b) Current total = 400,000 + 1,000 × 4,000 + 600,000 = **Rs. 5,000,000**; gap = **Rs. 200,000 above target** (Rs. 50 per unit). (c) Maximum materials = (4,800,000 − 400,000 − 600,000) ÷ 4,000 = **Rs. 950 per unit** (a cut of Rs. 50).

P8 · Equilibrium

(a) 200 − 5P = 40 + 3P gives 160 = 8P, so **P = 20**, Q = 200 − 100 = **100** (check: 40 + 60 = 100). (b) P = 25: Qd = 75, Qs = 115, **surplus 40** (price falls). (c) P = 15: Qd = 125, Qs = 85, **shortage 40** (price rises). (d) 260 − 5P = 40 + 3P gives 220 = 8P, so **P = 27.5, Q = 122.5** (check: 40 + 82.5 = 122.5). Price and quantity both rise.

P9 · Market table

(a) QD = QS = 500 at **Rs. 70, Q = 500**. (b) At Rs. 90: QD 100 vs QS 900, **surplus 800**. At Rs. 50: QD 900 vs QS 100, **shortage 800**. (c) New QS: 500, 700, 900, 1,100, 1,300. At Rs. 60: QD 700 = new QS 700, so the **new equilibrium is Rs. 60, Q = 700**: price falls by Rs. 10 and quantity rises by 200.

P10 · Shifts

	Curve	Direction	Price	Quantity
a	Supply	Right	Down	Up
b	Demand	Right	Up	Up
c	Demand for coffee (substitute)	Right	Up	Up
d	Supply	Left	Up	Down
e	Demand for sugary drinks	Left	Down	Down

P11 · PPC

(a) 10 tonnes of extra rice costs **10, 20, 30, 40** thousand nuts (100→90→70→40→0). (b) At 20 tonnes the maximum is 70 thousand nuts, so 80 is **unattainable** (outside the curve). (c) 50 is below 70, so the point is inside the curve: **inefficient** (resources unused or wasted; it is possible to produce more). (d) The PPC is concave: **increasing opportunity cost** (resources are not equally suited to both goods).

P12 · EPF and ETF

Employee EPF 8% = **Rs. 9,600**. Employer EPF 12% = **Rs. 14,400**. Employer ETF 3% = **Rs. 3,600**. Salary after EPF = 120,000 − 9,600 = **Rs. 110,400** (before any tax). Employer cost = 120,000 + 14,400 + 3,600 = **Rs. 138,000**.

P13 · Closing position

Item	Working	Closing
Cash	200,000 − 80,000 + 70,000 − 15,000 − 10,000	165,000
Inventory	80,000 − 40,000 (half sold at cost)	40,000
Total assets		**205,000**
Liabilities	none	**0**
Capital	200,000 + profit 30,000 (70,000 − 40,000) − rent 15,000 − drawings 10,000	**205,000**

Tally: 0 + 205,000 = 205,000. OK.

P14 · Mixed quick-fire

(a) 560 × 1.25 = **Rs. 700** (b) 975 ÷ 1.30 = **Rs. 750** (c) 360 = 18P, so **P = 20, Q = 240** (check: 40 + 200 = 240) (d) 120/50 = **2.4**; (120 − 45)/50 = **1.5** (e) 63/450 = **14%** (f) 54,000 × 365 ÷ 600,000 = 32.85, about **32.9 days** (g) 90/360 = **25%**

> **LAST REMINDER**
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> Formula → Substitution → Answer with units. Ratios වලට interpretation. Cost නොදුන්නොත් assumption. All the best for Nov 3!